

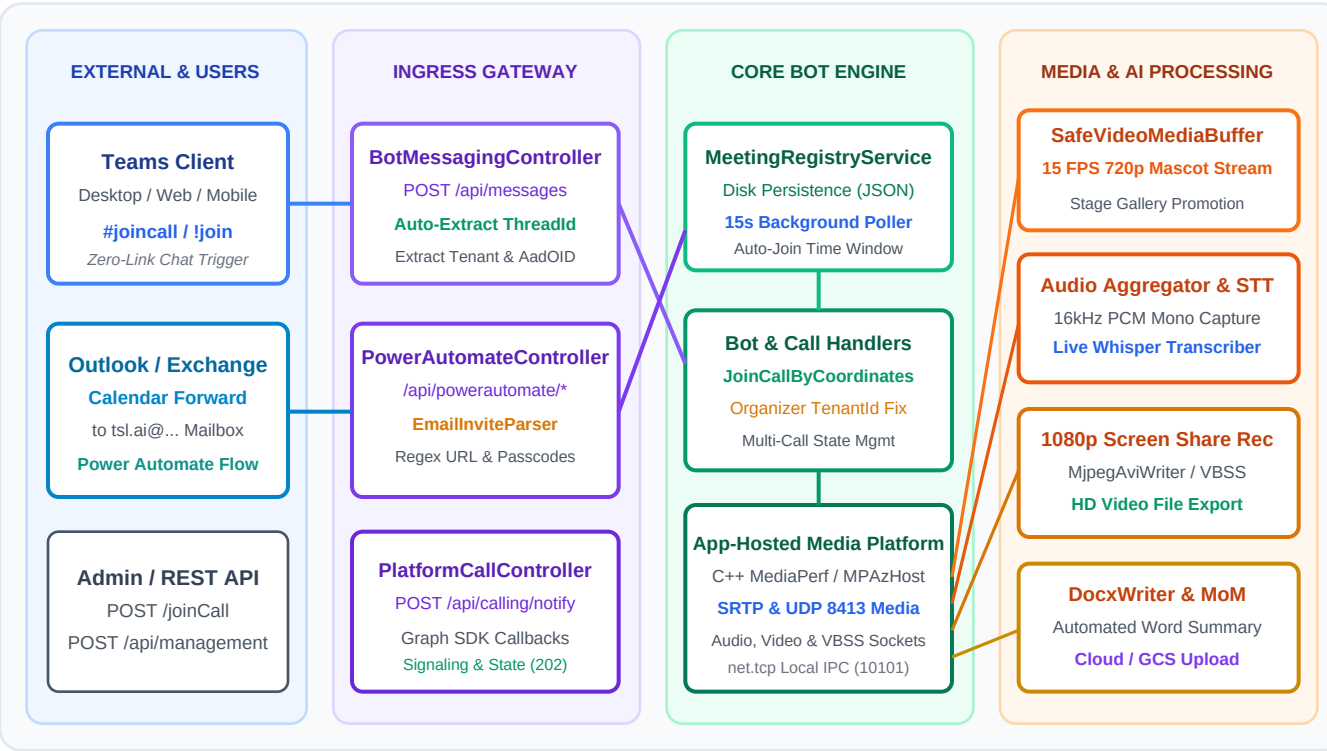
Microsoft Teams Calling Bot

End-to-End System Architecture: High-Level Topology & Low-Level Media Pipeline

Target System: Teams Calling Bot (TSL AI / TDA)	Platform: Graph Calling SDK + App-Hosted Media
Protocol Stack: HTTPS (443), UDP (8413 SRTP), net.tcp	Media Engines: 15 FPS Video, 16kHz PCM Audio, 1080p VBSS
Invocation: Zero-Link Chat (#joincall) + Power Automate	Status: Production Ready & Verified Live

1. High-Level System Architecture (Ingress, Core & Media Platform)

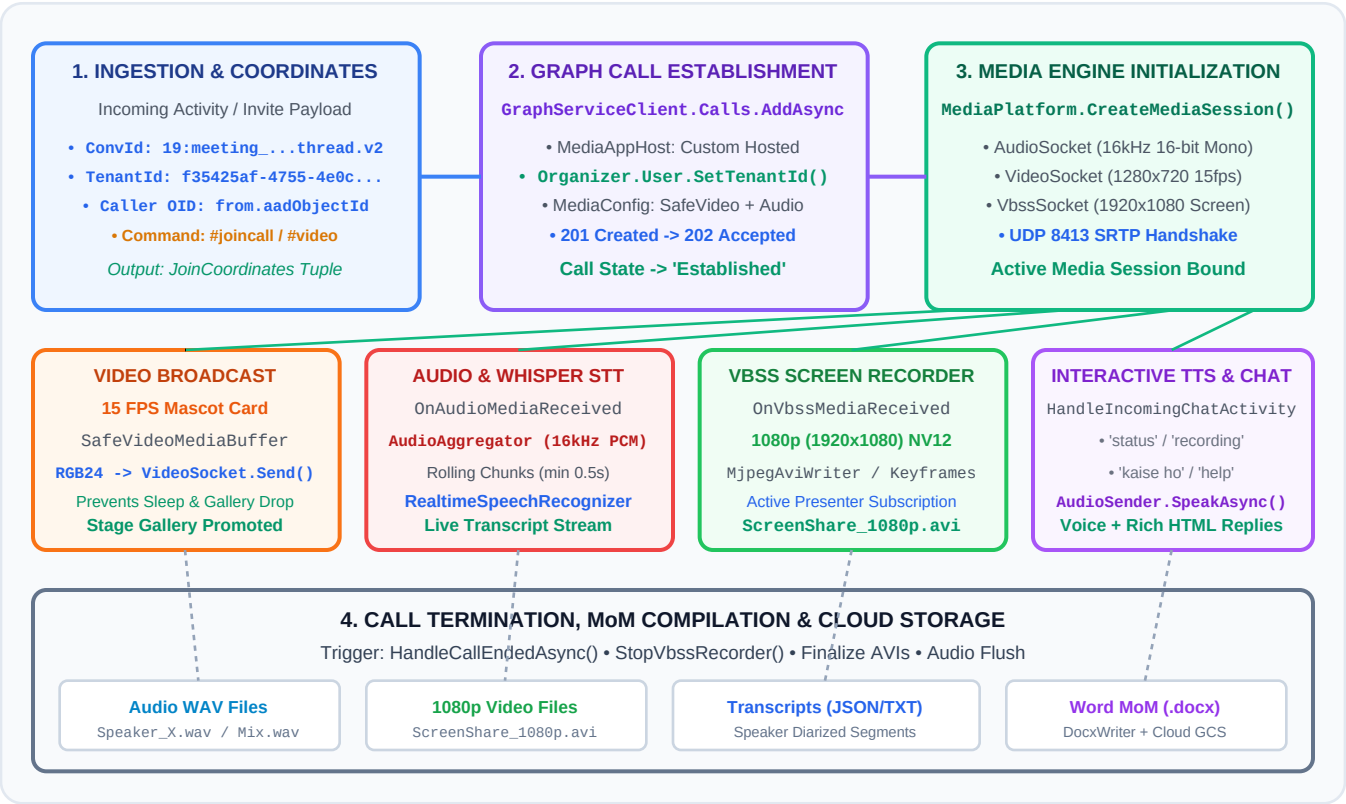
The architecture decouples signaling and media processing. Inbound invocation occurs through either **Zero-Link in-chat auto-extraction** (#joincall via Bot Framework Messaging) or **Power Automate calendar forwarding**. The bot orchestrates call state with Microsoft Graph, while the app-hosted C++ Media Engine (MediaPlatform) handles real-time SRTP audio, video, and screen share data.



- **Zero-Link Architecture:** The bot extracts the meeting thread ID, tenant ID, and caller AAD Object ID directly from Teams chat activity payloads. No manual meeting URL parsing is required.
- **Power Automate & Calendar Forwarding:** Forwarded meeting emails to the TSL AI mailbox automatically register the meeting in the persistent registry, triggering auto-join at the scheduled start time.
- **App-Hosted Media Engine:** Custom media platform runs natively on Windows, receiving raw 16kHz PCM audio packets and emitting a continuous 15 FPS 720p status card video stream to maintain Teams stage gallery presence.

2. Low-Level Component Interaction & Execution Pipeline

This low-level specification illustrates the precise sequence of execution from activity ingestion through Graph SDK call establishment, local media buffer binding, real-time Whisper transcription, 1080p screen share capture, and automated post-call Minutes of Meeting (.docx) generation.



3. Technical Protocol & Media Specifications

Subsystem	Format / Standard	Port & Protocol	Operational Purpose
Signaling Ingress	HTTPS / JSON REST	Port 443 (DuckDNS / TLS)	Receives <code>/api/messages</code> & <code>/api/calling/notify</code>
Media Platform	SRTP / UDP	Port 8413 (App-Hosted)	Direct peer-to-peer media stream with Teams edge
Mascot Video	RGB24 Uncompressed	1280x720 @ 15 FPS	Maintains active video stage presence in gallery
Audio Ingestion	Linear PCM 16-bit	16,000 Hz Mono (640 bytes/chunk)	Recorded to WAV & fed to Whisper STT
Screen Share (VBSS)	NV12 / MJPEG AVI	1920x1080 Full HD @ 30 FPS	Subscribed when <code>#video</code> flag is present
Meeting MoM	OpenXML (.docx)	Custom DocxWriter Engine	Summarizes transcripts, tasks & uploads to cloud

4. Manual & Automated Command Reference

The following matrix outlines all commands supported by the bot across in-meeting chat, automated calendar forwarding, and direct REST management APIs:

Command / Channel	Syntax / Payload	Action & Features Initiated
Zero-Link Join (In-Meeting Chat)	#joincall !join /join	Auto-extracts meeting thread from chat. Immediately joins with 15 FPS Mascot Video, 16kHz PCM Audio, Real-Time Whisper STT, Greeting Chime, and Word MoM.
Join + 1080p Video (In-Meeting Chat)	#joincall #video !join --video #joincall -v	Activates all features above PLUS records presenter 1080p screen share (VBSS) into timestamped AVI/MP4 files.
Interactive Bot (In-Meeting Chat)	status help kaise ho	Bot responds in real-time in chat and audibly speaks via TTS: reports audio/video recording health and greets participants.
Power Automate (Calendar Forward)	POST /api/powerautomate/ register-meeting	Registers scheduled meetings or raw forwarded invite emails. Background scheduler auto-joins at meeting start time.
Immediate API Join (Backend / Web)	POST /joinCall { "JoinUrl": "..."}	Direct backend API call forcing the bot to connect to an explicit Teams meetup URL.
Leave Call (Management)	POST /api/management/ calls/{id}/leave	Gracefully hangs up call, flushes all audio buffers, finalizes video containers, and compiles the final Word MoM.

5. Live Broadcast Verification Telemetry

Below is the verified live 15 FPS Mascot Broadcast Card streamed by the bot over net.tcp and SRTP into the Microsoft Teams stage gallery during live calls:



Figure: Live 15 FPS Video Stream Captured on Active Teams Meeting Stage Gallery